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Recurrent Laryngeal Nerve Regeneration Using a Self-Assembling Peptide Hydrogel

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Objectives/Hypothesis: To regenerate defected recurrent laryngeal nerves (RLNs), various methods have been developed. However, no consistently effective treatments are currently available because of their insufficient functional recovery. RADA16-I, a self-assembling peptide used clinically as a hemostat, reportedly supports neurite outgrowth and functional synapse formation in vitro. The purpose of this study was to investigate the effect of RADA16-I hydrogels on transected RLNs in rats.

Study Design: Animal experiments with controls.

Methods: Fifteen adult rats were divided into the following three groups: RADA16-I (+), RADA16-I (−), and neurectomy. A 6-mm gap of the left RLN was bridged using an 8-mm silicone tube in the RADA16-I (−) and RADA16-I (+) groups. Subsequently, RADA16-I hydrogel was injected into the tube in the RADA16-I (+) group. The surgical incisions were closed without any further treatment in the neurectomy group. After 8 weeks, laryngoscopy and electrophysiological and histological examinations were performed to evaluate the effect of RADA16-I on nerve regeneration and thyroarytenoid muscle atrophy.

Results: Although most rats in the three groups exhibited no improvements of their vocal fold movement, partial recovery was observed in one rat in the RADA16-I (+) group. The neurofilament-positive areas and the number of myelinated nerves in the RADA16-I (+) group were significantly higher than in the RADA16-I (−) group. The area of the left thyroarytenoid muscle in the RADA16-I (+) group was significantly larger than that of the neurectomy group.

Conclusions: Our results suggested that RADA16-I hydrogel was effective for RLN regeneration.

Key Words: Vocal fold, paralysis, RADA16-I, peripheral nerve, regeneration.

Level of Evidence: NA

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INTRODUCTION

The recurrent laryngeal nerve (RLN) is a branch of the vagus nerve that terminates at the muscles in the neck, including the laryngeal muscles, and controls vocal fold (VF) movements. Injury to the RLN can result in VF paralysis that can in turn lead to dysphonia, airway obstruction, and possible aspiration; therefore, this type of injury can severely impair an individual's quality of life.[1,2] Surgical iatrogenic injuries associated with the removal of thyroid tumors are the most common cause of VF immobility.[2] Chiang et al. reported that RLN invasion was identified in 19.4% of patients with thyroid cancer.[3] RLNs invaded by tumors require excision; therefore, RLN defects occur in such patients. Although various methods to regenerate defected RLNs have been reported,[4–11] no consistently effective treatments are currently available because they lead to insufficient functional recovery and misdirected innervation.[12] At present, preventing laryngeal muscle atrophy by maintaining nerve continuity is the only feasible treatment method for injured RLNs.[13]

Once neurons become damaged following trauma, inflammation, or tumors, or from surgical treatments, such as the combined resection of tumors and peripheral nerves, permanent loss of neural function may occur. Although the axonal regeneration capacity in the peripheral nervous system is greater than in the central nervous system, because the responses of the respective glial cells to injury are different,[14] spontaneous peripheral nerve repair remains challenging because of poor functional recovery.[15]

The first-choice treatment for injured peripheral nerves is end-to-end anastomosis. If the nerve defect is very large, however, autologous transplantation is the most common treatment. Unfortunately, autologous transplantation can sometimes lead to other neural disorders, including neuroma, in the donor sites. Additionally, the size difference between the donor and recipient nerves can be problematic.[1,16,17]

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To resolve these problems, various devices that can substitute the injured parts of nerves have been developed.[16] These devices may be made of plastic or biological materials and are used to redirect axon elongation away from the uninjured part of a nerve. For instance, silicone (polydimethylsiloxane) tubes, encapsulating adipose-derived stem cells and inserted between nerve stumps, reportedly induced regeneration in rat sciatic nerves.[18] A bioabsorbable polyglycolic acid tube filled with collagen sponge enabled the reconnection of canine peroneal nerves.[19] Such tubes have also been investigated in clinical studies and were reported to improve functional nerve regeneration.[20,21] With regard to treatments for defected RLNs, various types of nerve conduits have been investigated using animal models,[4–11] in which both the polyglycolic acid tubes[4,11] and silicone tubes[5] have been used. It has also been reported that a polyglycolic acid tube filled with collagen sponge was used as a scaffold for nerve regeneration.[7] In the present study, we focused on a self-assembling peptide (SAP) as the material with which to fill such tubes, in a similar way as a collagen sponge. A SAP is a synthetic chemical that can aggregate into nanofibers under physiological conditions. SAPs have been developed as a biomaterial for use in overcoming existing challenges in various medical fields, including regenerative medicine. SAPs are highly versatile because of their unique properties, such as their ability to encapsulate cells,[22] high water content,[23] ease of modification,[24] low toxicity, and biocompatibility.[25,26] SAPs are completely synthetic chemicals, unlike other hydrogels, which are made of animal-derived biomaterial[27]; therefore, SAP hydrogels are considered to be highly suitable for clinical applications and have already been clinically used as hemostats.[28]

One of the extensively utilized SAPs is RADA16-I, which consists of 16 alternative amino acids (hydrophilic arginine, hydrophobic alanine, hydrophilic aspartic acid). This amphiphilic peptide can form a hydrogel when in an environment with a neutral pH.[23,29] The amino acids in RADA16-I are constituents of the human body, and RADA16-I hydrogel seems to possess similar properties to the extracellular matrix (ECM).[30,31] An in vitro study using the neuronal lineage cell PC12 has suggested that RADA16-I hydrogel supports neurite outgrowth and synapse formation by acting as a scaffold.[25] An in vivo study using a rat model of injured sciatic nerves showed that RADA16-I hydrogel provided nanofiber microenvironments and promoted nerve regeneration.[32]

The aim of this study was to investigate the efficacy of RADA16-I hydrogel for RLN regeneration. We implanted silicone tubes filled with RADA16-I into rat RLN gaps and examined the tubes' therapeutic effects on the RLN defect, VF paralysis, and thyroarytenoid (TA) muscle atrophy.

MATERIALS AND METHODS

Animals

Fifteen 10-week-old male Sprague-Dawley rats were used in this study. All rats were purchased from Japan SLC (Shizuoka, Japan). The rats were randomly divided into the following groups: RADA16-I (+) (n = 5), RADA16-I (−) (n = 5), and neurectomy (n = 5) groups. Animals were cared for in the Institute of Laboratory Animals of the Graduate School of Medicine, Kyoto University, Japan. All experimental procedures were performed under general anesthesia, and the study was approved by the Animal Research Committee of Kyoto University.

Preparation of the RADA16-I Hydrogel Solution

RADA16-I was provided by 3-D Matrix (Tokyo, Japan) and was produced according to Good Manufacturing Practice guidelines. RADA16-I consists of 16 alternative amino acids, and the amphiphilic peptide can form a hydrogel when its environment has a neutral pH. The RADA16-I was stored at 4°C prior to use. A 1% (w/v) RADA16-I solution was prepared by diluting the RADA16-I in autoclaved Milli-Q water in a microtube, then Dulbecco's Modified Eagle Medium (Nacalai Tesque, Kyoto, Japan) was added, and the mixture was stirred to make a RADA16-I hydrogel solution. All procedures were conducted under sterile conditions. The final concentration of RADA16-I in the solution was 5 mg/mL. Neutral pH was confirmed using pH test paper.

Implantation of RADA16-I Hydrogel

General anesthesia was induced by an intraperitoneal injection of butorphanol tartrate (0.25 mg/100 g), medetomidine hydrochloride (0.015 mg/100 g), and midazolam (0.2 mg/100 g). The left RLN was separated from the surrounding tissue and resected to create a 6-mm gap (Fig. 1A). In the neurectomy group, the surgical incisions were closed without any further treatment. In the other groups, an 8-mm silicone tube (inner diameter = 1 mm) was sutured between the nerve stumps with 10-0 nylon at one point each on the cranial and caudal sides (Fig. 1B). Both ends of the nerve stumps (up to 1 mm) were inserted into the silicone tube. RADA16-I hydrogel solution (7 μL) was subsequently injected into the silicone tube in the RADA16-I (+) group (Fig. 1C). In the RADA16-I (−) group, no RADA16-I hydrogel was injected. All rats survived the operation and no complications were observed during the experimental period.

Laryngoscopy

Bilateral movement of VFs before the operation and left VF paralysis following RLN transection were confirmed using a rigid endoscope (Olympus A70961A; Olympus, Tokyo, Japan). Eight weeks after the operation, laryngoscopy was performed again to observe VF movement. In previous studies that used a rat injured-RLN model, 12 or 15 weeks postoperatively was selected as the experimental period.[5,7,8,10] Our preliminary studies suggested that nerve regeneration from proximal to distal stumps could be confirmed by histological examination 8 weeks postoperatively, so this was the experimental period selected to investigate the relatively earlier effect of using RADA16-I hydrogel with a rat injured-RLN model rather than the experimental period described in previous reports.

Electrophysiological Examination

The RLN and larynx were exposed, and an electrophysiological examination was performed. Rubber material was placed under the left RLN at a 2-mm site proximal to the resected region and under the right RLN at a similar position. The nerves were stimulated using stimulation electrode needles, which were placed on the nerves at the sites overlaying the rubber. Myograms were recorded using a Neuropack2 (MEB-5100; Nihon Kohden, Tokyo, Japan). The compound muscle action potential (CMAP) amplitude ratio of the treated side to the untreated side was calculated.

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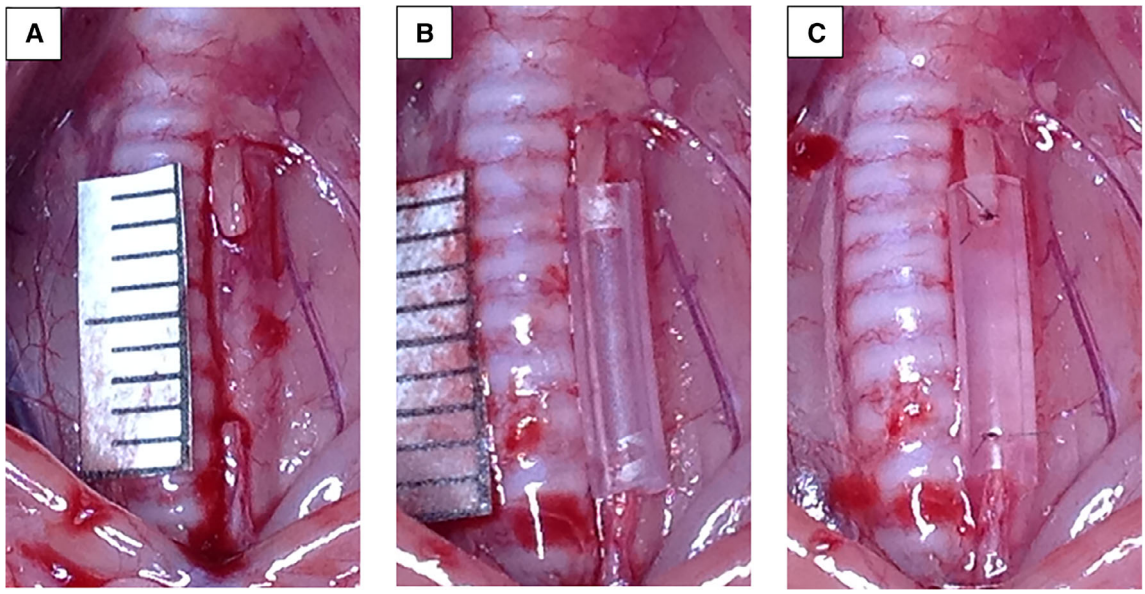


Fig. 1. Implantation of RADA16-I hydrogel to a resected recurrent laryngeal nerve (RLN). The left RLN was resected to create a 6-mm gap (A). The nerve stumps were sutured with 10-0 nylon, and up to 1 mm of the nerve was inserted into the silicone tube (B). Subsequently, 7 μL RADA16-I hydrogel solution was injected into the silicone tube in rats of the RADA16-I (+) group (C).

Immunohistochemical and Histological Examination

CO_2 euthanasia was performed following laryngoscopy and electrophysiological examination. The larynx and trachea were removed, along with the RLNs. The larynx was fixed in 4% paraformaldehyde in phosphate-buffered saline (PBS) overnight at 4°C, prior to use for the evaluation of TA muscle atrophy. Regenerated RLNs were divided into four 1.5-mm-long segments and used to evaluate axon growth and myelination.

Immunohistochemistry of Regenerated Nerves to Evaluate the Axon Area

A regenerated nerve was divided into four equal segments. The third segment from the proximal side was immersed in PBS containing 30% sucrose for cryoprotection and embedded in optimum cutting temperature compound. Transverse cryosections were made at a thickness of 10 μm. The center of a regenerated nerve (proximal end of the segment) was used for immunohistochemistry (see Supporting Figure S1 in the online version of this article). After rinsing with PBS, the sections were treated with 0.2% Triton X-100 in PBS for 30 minutes at 4°C to allow permeabilization. To block nonspecific antibody binding, sections were immersed in 2% bovine serum albumin in PBS for 1 hour at 4°C. The sections were then incubated with rabbit polyclonal anti-200-kDa neurofilament (NF-200) antibody (1:800; Sigma-Aldrich, St. Louis, MO) overnight at 4°C. Alexa Fluor 488 donkey anti-rabbit immunoglobuline G (1:1000; Invitrogen, Carlsbad, CA) was used as a secondary antibody. After further washing with PBS, images were obtained using a fluorescence microscope (Olympus BX 50; Olympus). The NF-200–positive area was measured using ImageJ software (National Institutes of Health, Bethesda, MD).

Toluidine Blue Staining to Count the Number of Myelinated Nerves

Paraformaldehyde-fixed nerves were further fixed with 2.5% glutaraldehyde, then with 1% osmic acid, before being embedded in epoxy resin. Transverse semithin sections (1-μm thickness) were obtained from the proximal, middle, and distal portions of the regenerated nerves, with intervals of approximately 1.5 mm. The sections were stained with 0.5% (w/v) toluidine blue solution, and images were obtained by light microscopy (Nikon ECLIPSE 80i; Nikon, Tokyo, Japan). The total number of myelinated nerves was counted at the proximal, middle, and distal sections.

Transmission Electron Microscopy for the Morphological Evaluation of Myelinated Nerves

The epoxy resin–embedded nerves were also observed for transmission electron microscopy (TEM). Transverse ultrathin sections (70 nm) were obtained from the middle portion of the regenerated nerves (see Supporting Figure S1 in the online version of this article). The sections were stained with uranyl acetate and lead citrate, then observed using a transmission electron microscope (Model H-7000; Hitachi High-Technologies, Tokyo, Japan).

Histological Examination to Evaluate Thyroarytenoid Muscle Atrophy

The paraformaldehyde-fixed larynx was immersed in PBS containing 30% sucrose overnight and embedded in optimal cutting temperature compound. Coronal cryosections (10-μm thick) were obtained from the anterior, middle, and posterior portions of the VF, with intervals of 250 μm. The inclination of the sections was adjusted as described in the Supporting Figure S2 and Supplemental Methods in the online version of this article. Hematoxylin and eosin staining was carried out, and the area of the TA muscle was measured using ImageJ software. The percentages of the left TA muscle area relative to the right TA muscle area were calculated for the respective sections. These percentages of the respective sections were averaged in individual rats and compared between the neurectomy, RADA16-I (–), and RADA16-I (+) groups.

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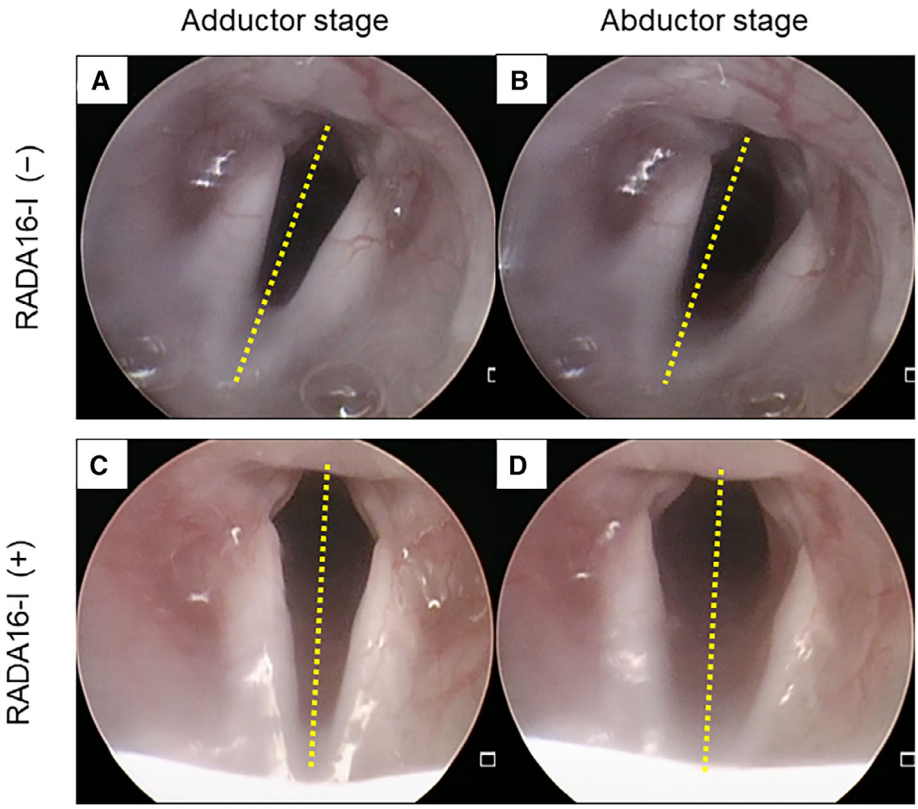


Fig. 2. Endoscopic findings of vocal fold movement 8 weeks postoperatively. No improvements were identified in the RADA16-I (–) group (A, B). Partial recovery of left vocal fold movement was observed in one rat in the RADA16-I (+) group (C, D). The midline is indicated by a yellow dotted line.

Statistical Analysis

All statistical analyses were performed using JMP 14.0.0 software (SAS Institute, Cary, NC). Wilcoxon rank sum tests were performed to analyze differences in the CMAP amplitude ratio, size of the NF-200–positive area, and number of myelinated fibers between the RADA16-I (–) and RADA16-I (+) groups. The Steel-Dwass test was carried out to compare TA muscle areas in the neurectomy, RADA16-I (–), and RADA16-I (+) groups. Differences were considered statistically significant at a p value of <0.05.

RESULTS

Laryngoscopy

Bilateral movements of the VF were observed in all rats prior to RLN transection, and the left VF was paralyzed in all rats following RLN transection. Eight weeks after the operation, no improvements in VF movement were identified in the rats of the RADA16-I (–) and neurectomy groups (Fig. 2A,B and Movie 1). Although most rats in the RADA16-I (+) group did not exhibit any improvements either, partial recovery was observed in one rat in the RADA16-I (+) group (Fig. 2C,D and Movie 2). The abduction of the left VF was weaker than the right VF in this rat. However, consistent movement of the left VF was observed.

Electrophysiological Examination

The CMAP amplitude was detectable in all treated nerves except for one in the RADA16-I (–) group. Although the average of the CMAP amplitude ratio in the RADA16-I (+) group was slightly higher than that in the RADA16-I (–) group, there was no significant difference between the groups (Fig. 3). In the one rat that showed partial recovery of VF movement, an equivalent CMAP amplitude was measured between the untreated and treated sides.

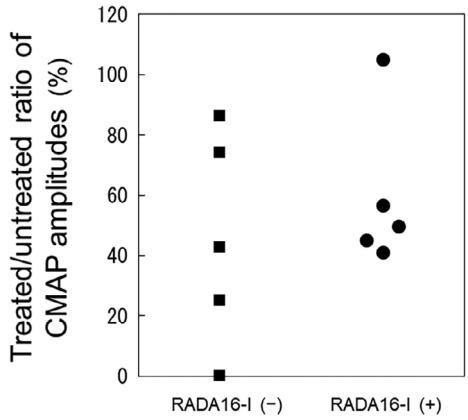


Fig. 3. The treated/untreated ratio of compound muscle action potential (CMAP) amplitudes. The CMAP amplitude ratio in the RADA16-I (–) and RADA16-I (+) groups is shown as a dot plot.

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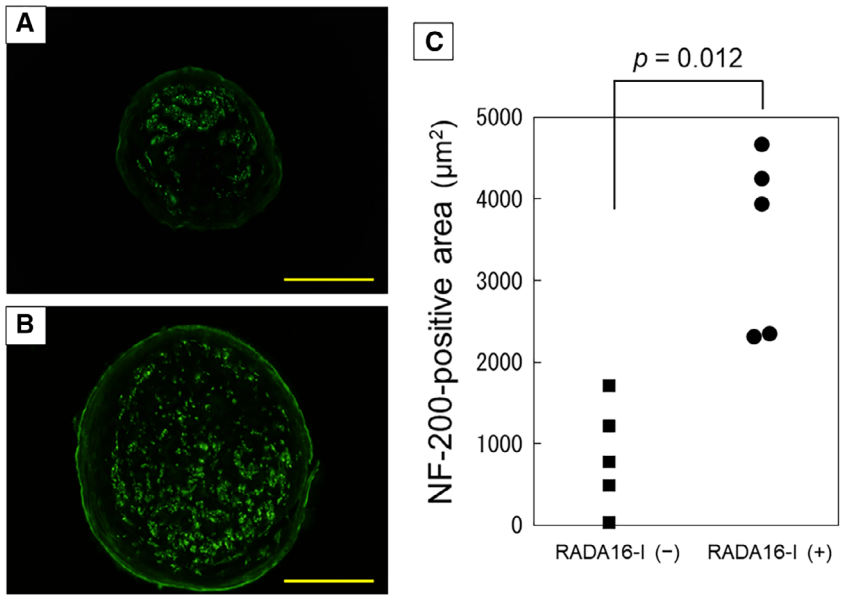


Fig. 4. Immunohistochemistry of NF-200 expression. The regenerated nerve in the RADA16-I (–) (A) and RADA16-I (+) groups (B). Bar = 100 μm. The NF-200-positive areas of the RADA16-I (+) group are shown as a dot plot (C).

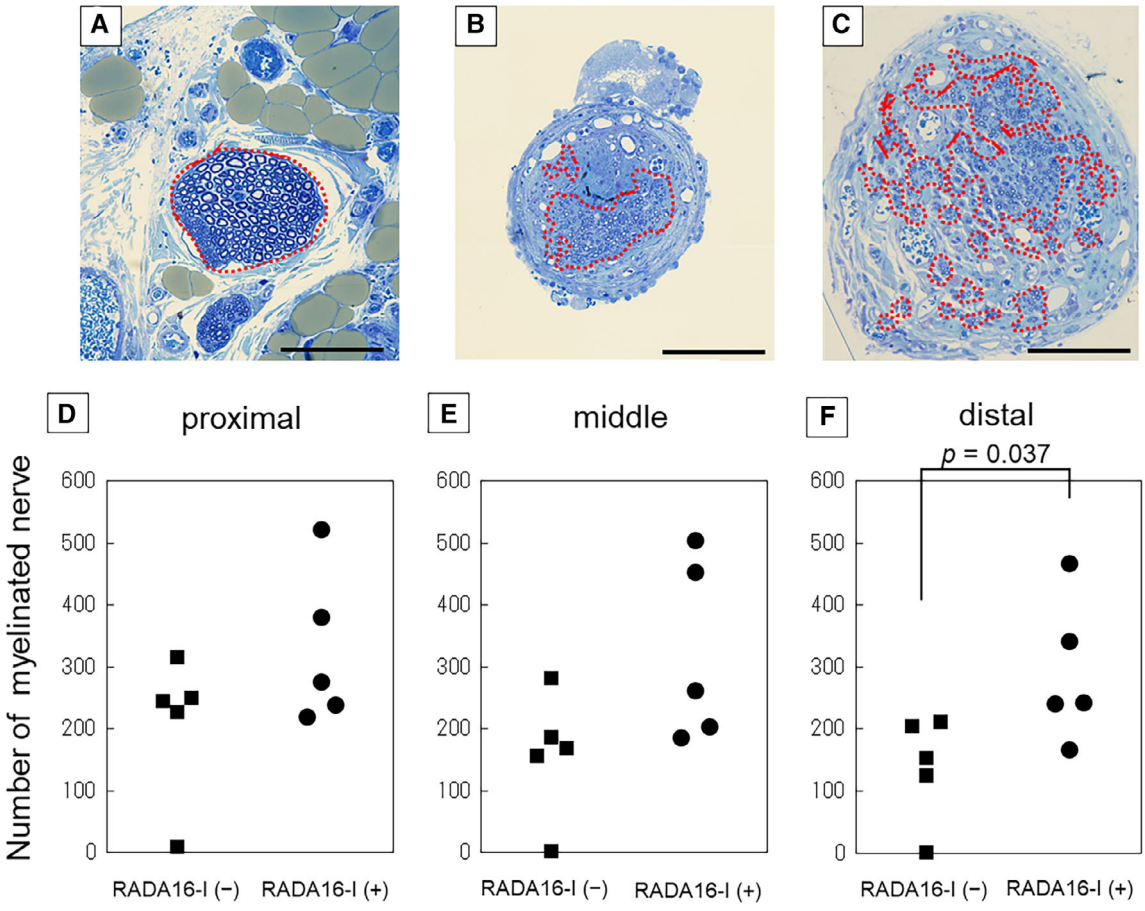


Fig. 5. Toluidine blue staining and the number of myelinated nerves. Untreated recurrent laryngeal nerve (A). Middle portion of regenerated nerves in the RADA16-I (–) group (B) and the RADA16-I (+) group (C). Myelinated nerves are identified inside the red dotted line. Bar = 100 μm. The numbers of myelinated nerves at the proximal (D), middle (E), and distal (F) portions are shown as dot plots.

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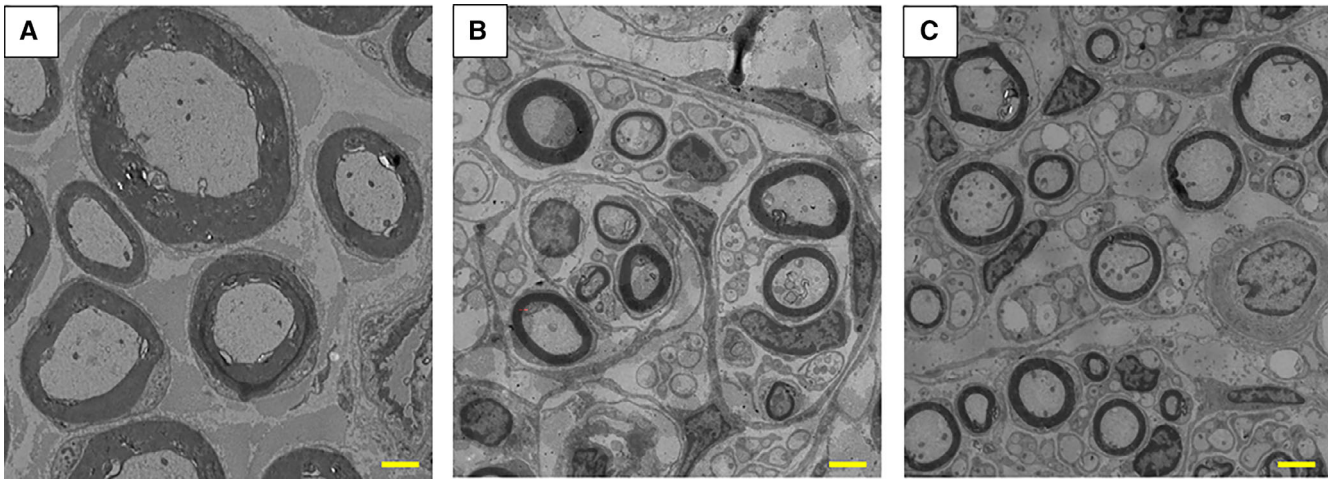


Fig. 6. Myelinated nerves observed by transmission electron microscopy. Images obtained of untreated recurrent laryngeal nerves (A) and regenerated nerves in the RADA16-I (–) (B) and RADA16-I (+) (C) groups are shown. Bar = 2 μm.

NF-200 Expression Area

Eight weeks after the operation, continuous tissues were macroscopically observed inside the silicone tubes in the RADA16-I (+) and RADA16-I (–) groups, whereas no continuous tissues were observed between the proximal and distal stumps in the neurectomy group. Neurofilaments are thought to be the main constituent of axons, and NF-200 is mainly found in mature neurons.[33,34] An increased number of sites with positive NF-200 staining were detected in the RADA16-I (+) group compared with the RADA16-I (–) group (Fig. 4A,B), and the NF-200–positive area of the RADA16-I (+) group was significantly larger than that of the RADA16-I (–) group ($p = 0.012$) (Fig. 4C).

Regeneration of Myelinated Nerves

In the rat of the RADA16-I (–) group where the CMAP amplitude was not measured, the regeneration of myelinated nerves was restricted to the proximal portion of the continuous tissue. However, in the remaining rats, myelinated nerves were observed to extend from the proximal to the distal portions. Myelinated nerves were

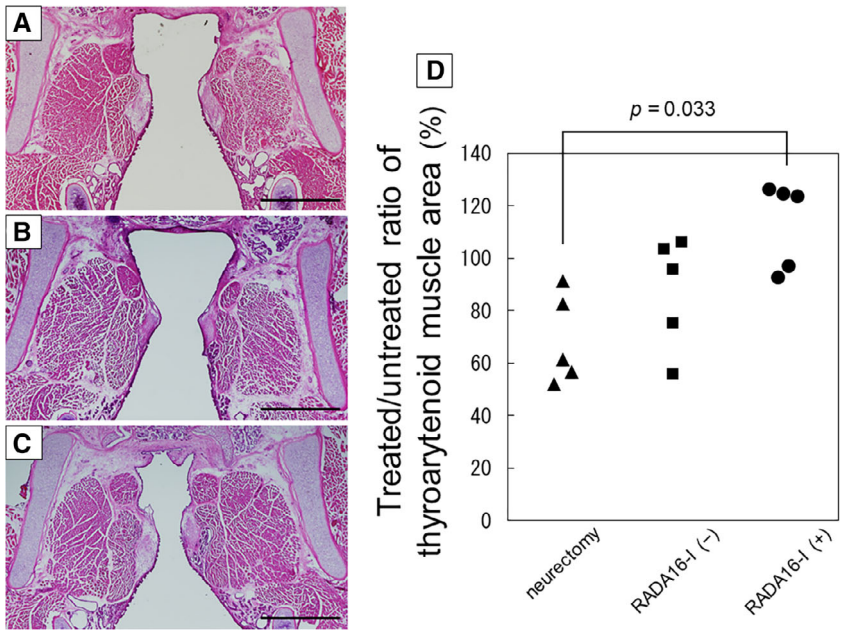


Fig. 7. Evaluation of thyroarytenoid muscle atrophy by hematoxylin and eosin staining in the neurectomy (A), RADA16-I (–) (B), and RADA16-I (+) (C) groups. Bar = 1 mm. The treated/untreated ratio of the thyroarytenoid muscle area is shown as a dot plot.

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ubiquitously distributed in the continuous tissue of the RADA16-I (+) group, whereas this pattern was not observed in the RADA16-I (–) group (Fig. 5A–C). The number of myelinated nerves at the distal portion in the RADA16-I (+) group was significantly higher than that in the RADA16-I (–) group ($p = 0.037$); however, there was no significant difference between the groups at the proximal and middle portions (Fig. 5D–F). TEM confirmed the myelination of regenerated nerves in the RADA16-I (+) and RADA16-I (–) groups. There were no observable differences in the thickness of the myelin sheath or the axon diameters between the nerves in the RADA16-I (+) and RADA16-I (–) groups (Fig. 6).

TA Muscle Atrophy

The area of the left TA muscle in the neurectomy group decreased to approximately 70% of that of the right TA muscle (Fig. 7). TA muscle atrophy was not prevented in the RADA16-I (–) group; however, the area of the left TA muscle in the RADA16-I (+) group was significantly higher than that of the neurectomy group ($p = 0.033$).

DISCUSSION

Nerve branches that input into the abductor and adductor muscles coexist in the RLN. Reconnection of a transected RLN can result in the misdirection of these branches. In our study, we showed that in most cases, VF movement could not be improved. This result can be attributed to misdirection. One rat in the RADA16-I (+) group exhibited partial recovery of its VF movement. Interestingly, the highest number of myelinated nerves was observed in this rat. It has been suggested that additional inputs from the superior laryngeal nerve may complement the role of the RLN following disruption of the RLN input[35]; hence, it could not be concluded that the recovery of VF movement was due to RLN regeneration.

Our immunohistochemical and histological studies suggested that RADA16-I promoted axonal growth and myelination in the RLN defects. In a previous in vitro study, RADA16-I was shown to support the cell adhesion, neurite outgrowth, and synapse formation in PC12 cells and rat hippocampal neurons.[24] Therefore, the regenerative effect of RADA16-I observed in our study might be reliant on the promotion of neural cell adhesion and neurite elongation from the RLN stump. This hypothesis is supported by other in vivo studies, which showed that an injection of RADA16-I promoted axon regeneration and the return of functional vision in a hamster model of optic-tract cuts,[27] and accelerated nerve healing in a rat model of sciatic nerve injuries by providing nanofiber microenvironments.[32] RADA16-I is considered useful as the composite of a nerve conduit because of its ability to provide a microenvironment that promotes axonal growth and reconnects distant nerve stumps. Furthermore, there has been speculation that RADA16-I promotes nerve regeneration by inhibiting the invasion of scar tissue.[36]

In this study, the ratio of left TA muscle area to right TA muscle area in the RADA16-I (+) group (but not in the RADA16-I (–) group) was significantly higher than that in the neurectomy group. Our results imply that RADA16-I is useful for preventing laryngeal muscle atrophy by promoting RLN regeneration. However, Pitman et al.[37] have shown that maturation of reinnervation requires a minimum of 16 weeks after connecting resected nerves using saline-hydrated gel foam. Here we only evaluated nerve regeneration for 8 weeks postoperatively; therefore, long-term studies may be necessary to confirm the effect of RADA16-I on nerve regeneration.

To repair large gaps in peripheral nerves, Bellamkonda[38] advocated the following essential requisites: 1) hydrogels or fibers to act as frameworks that promote growth, 2) neurostimulatory ECM, 3) neurotrophic factors, and 4) glial cells or other supporting cells such as Schwann cells or stem cells. RADA16-I can fulfill the first two requirements because its three-dimensional structure acts as a scaffold for cell growth and it shows similarity to the ECM.[30,31] The administration of neurotrophic factors in conjunction with RADA16-I,[39] and the encapsulation of Schwann cells[40]or mesenchymal stem cells[22] by RADA16-I, may promote peripheral nerve regeneration by overcoming the remaining conditions. In addition, by attaching functional motifs to RADA16-I, this hydrogel is able to yield various features.[24] The RGD (arginine-glycine-aspartate) motif is found within several ECM proteins where it serves as a binding site for some cell adhesion receptors[41] and acts as a pathfinding source during axon elongation.[42] Although our results suggest that RADA16-I itself is beneficial for RLN regeneration, it may be possible that modification with functional motifs could further improve the regenerative effect of RADA16-I.

CONCLUSION

RADA16-I is a feasible material because of its biocompatibility and versatility, which has already been used clinically as a hemostat. Our histological results indicated that RADA16-I hydrogel improved RLN regeneration and prevented TA muscle atrophy in a rat model of RLN injury, indicating the usefulness of RADA16-I for the therapeutic treatment of RLN defects.

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